CHE 201 HW 7

Due: 11.59pm on 10/26/2019

1. Refrigerant 134A enters a well-insulated nozzle at 200 lbf/in^2, 220 ^oF, with a velocity of 120 ft/s and exits at 20 lbf/in^2 with a velocity of 1500 ft/s. For steady state operation and neglecting potential energy effects:

a) Draw a schematic of the system and label the system boundary.
b) State the engineering model
c) Write the simplified mass rate and energy rate balance.
d) Determine the exit temperature in ^oF
e) Carefully sketch the process on a P-ν diagram

2. Air expands through a turbine from 8 bar, 960 K to 1 bar, 450 K. The inlet velocity is small compared to the exit velocity of 90 m/s. The turbine operates at steady state and develops a power output of 2500 kW. Heat transfer between the turbine and its surroundings and potential energy effects are negligible. Modeling air as an ideal gas:

a) Draw a schematic of the system and label the system boundary.
b) State the engineering model
c) Write the simplified mass rate and energy rate balance.
d) Calculate the mass flow rate of the air, in kg/s, and the exit area, in m^2.
e) Carefully sketch the process on a T-ν diagram

3. Water vapor enters a turbine operating at steady state at 500^oC, 40 bar, with a velocity of 200 m/s, and expands adiabatically to the exit, where it is a saturated vapor at 0.8 bar, with a velocity of 150 m/s and a volumetric flow rate of 9.48 m^3/s. Neglecting potential energy effects:

a) Draw a schematic of the system and label the system boundary.
b) Carefully sketch the process on a P-ν diagram
c) State the engineering model
d) Write the simplified mass rate and energy rate balance.
e) Determine the power developed by the turbine, in kW.

4. Refrigerant 134A enters an air conditioner compressor at 4 bar, 20°C, and is compressed at steady state to 12 bar, 80 °C. The volumetric flow rate of the refrigerant entering is 4 m^3/min. The work input to the compressor is 60 kJ per kg of refrigerant flowing. Neglecting kinetic and potential energy effects:

a) Draw a schematic of the system and label the system boundary.
b) Carefully sketch the process on a T-ν diagram
c) State the engineering model
d) Write the simplified mass rate and energy rate balance.
e) Determine the heat transfer rate, in kW

5. A pump at steady state is drawing water at a temperature of 15 °C and a pressure of 1 bar from a reservoir and delivering it at a pressure of 3 bar to a storage tank that is 15 m above reservoir. The mass flow rate of the water is 1.5 kg/s. The water temperature remains nearly constant at 15 °C, and there is no significant change in kinetic energy from inlet to exit. Heat transfer between the pump and the surroundings is negligible.

a) Draw a schematic of the system and label the system boundary.
b) State the engineering model
c) Write the simplified mass rate and energy rate balance.
d) Determine the power required by the pump, in kW. Let $g = 9.81$ m/s^2

6. Ammonia enters the expansion valve of a refrigeration system at a pressure of 10 bar, and temperature of 24 °C, and exits a 1 bar. If the refrigerant undergoes a throttling process:

a) Draw a schematic of the system and label the system boundary.
b) State the engineering model
c) Write the simplified mass rate and energy rate balance.
d) Determine the quality of the refrigerant exiting the expansion valve.
e) Carefully sketch the process on a T-ν diagram